```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
df = pd.read_csv("/Sample - Superstore.csv", encoding="ISO-8859-1")
print(df. head())
```

```
   Row ID        Order ID  Order Date   Ship Date       Ship Mode Custo
0       1  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CO
1       2  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CO
2       3  CA-2016-138688   6/12/2016   6/16/2016    Second Class    DV
3       4  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO
4       5  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Postal Code  Region       Product ID         Category Sub-Category  \
0       42420   South  FUR-BO-10001798        Furniture    Bookcases
1       42420   South  FUR-CH-10000454        Furniture       Chairs
2       90036    West  OFF-LA-10000240  Office Supplies       Labels
3       33311   South  FUR-TA-10000577        Furniture       Tables
4       33311   South  OFF-ST-10000760  Office Supplies      Storage

                                        Product Name     Sales  Quantit
0                 Bush Somerset Collection Bookcase  261.9600
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
4                    Eldon Fold 'N Roll Cart System   22.3680

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164

[5 rows x 21 columns]
```

```
print("Dataset shape:", df.shape)

print("\nColumn names:")
print(df.columns.tolist())

print("\nDataset information:")
df.info()
```

```
print("\nStatistical summary:")
display(df.describe())

print("\nFirst 5 rows:")
display(df.head())
```

```
Dataset shape: (9994, 21)

Column names:
['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Custo

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB

Statistical summary:
```

	Row ID	Postal Code	Sales	Quantity	Discount	
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	999
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	2
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	23
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-659
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	2
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	839

First 5 rows:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segm
	CA-						

```
print(df.isnull().sum())
```

```
Row ID           0
Order ID         0
Order Date       0
Ship Date        0
Ship Mode        0
Customer ID      0
Customer Name    0
Segment          0
Country          0
City             0
State            0
Postal Code      0
Region           0
Product ID       0
Category         0
Sub-Category     0
Product Name     0
Sales            0
Quantity         0
Discount         0
Profit           0
dtype: int64
```

```
print("Duplicate rows:", df.duplicated().sum())
df = df.drop_duplicates()

print("Duplicates remaining:", df.duplicated().sum())
```

```
Duplicate rows: 0
Duplicates remaining: 0
```

```
df["Sales"] = df["Sales"].fillna(df["Sales"].median())
```

```
df["Category"] = df["Category"].fillna("Unknown")
```

```
print(df.head())
print(df.info())
print(df.isnull().sum())
```

```
   Row ID        Order ID  Order Date   Ship Date       Ship Mode Cus
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[5 rows x 21 columns]
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 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
```

```
total_sales = df["Sales"].sum()

total_profit = df["Profit"].sum()

total_quantity = df["Quantity"].sum()

total_orders = df["Order ID"].nunique()

profit_margin = (
    (total_profit / total_sales) * 100
    if total_sales != 0
    else 0
)

print("========== SALES KPIs ==========")

print(f"Total Revenue: ${total_sales:,.2f}")

print(f"Total Profit: ${total_profit:,.2f}")
```

```
print(f"Total Quantity Sold: {total_quantity:,}")

print(f"Total Orders: {total_orders:,}")

print(f"Profit Margin: {profit_margin:.2f}%")
```

```
========== SALES KPIs ==========
Total Revenue: $2,297,200.86
Total Profit: $286,397.02
Total Quantity Sold: 37,873
Total Orders: 5,009
Profit Margin: 12.47%
```

```
category_sales = (
    df.groupby("Category")["Sales"]
    .sum()
    .sort_values(ascending=False)
)

print("Sales by Category:")
display(category_sales.reset_index())
```

```
Sales by Category:
```

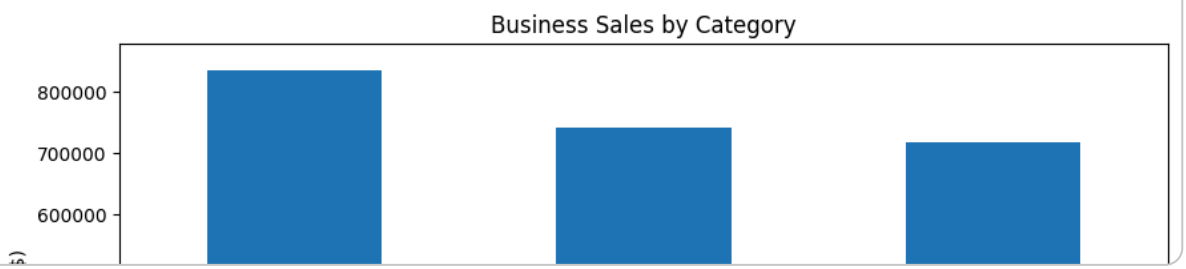
	Category	Sales
0	Technology	836154.0330
1	Furniture	741999.7953
2	Office Supplies	719047.0320

```
plt.figure(figsize=(9, 5))

category_sales.plot(kind="bar")

plt.title("Business Sales by Category")
plt.xlabel("Product Category")
plt.ylabel("Total Sales ($)")

plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```




```
top_products = (
    df.groupby("Product Name")["Sales"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)
```

```
print("Top 10 Products by Revenue:")
display(top_products.reset_index())
```

```
Top 10 Products by Revenue:
```

	Product Name	Sales
0	Canon imageCLASS 2200 Advanced Copier	61599.824
1	Fellowes PB500 Electric Punch Plastic Comb Bin...	27453.384
2	Cisco TelePresence System EX90 Videoconferenci...	22638.480
3	HON 5400 Series Task Chairs for Big and Tall	21870.576
4	GBC DocuBind TL300 Electric Binding System	19823.479
5	GBC Ibimaster 500 Manual ProClick Binding System	19024.500
6	Hewlett Packard LaserJet 3310 Copier	18839.686
7	HP Designjet T520 Inkjet Large Format Printer ...	18374.895
8	GBC DocuBind P400 Electric Binding System	17965.068
9	High Speed Automatic Electric Letter Opener	17030.312

```
region_sales = (
    df.groupby("Region")["Sales"]
    .sum()
    .sort_values(ascending=False)
)

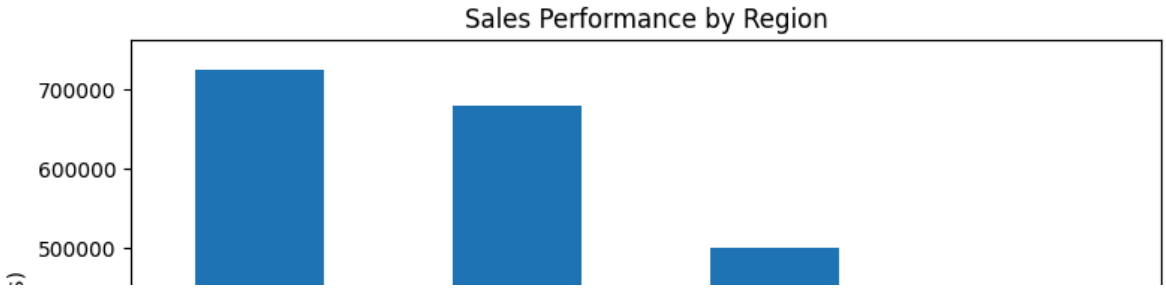
display(region_sales.reset_index())
```

```
plt.figure(figsize=(8, 5))
region_sales.plot(kind="bar")

plt.title("Sales Performance by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales ($)")
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()
```

	Region	Sales
0	West	725457.8245
1	East	678781.2400
2	Central	501239.8908
3	South	391721.9050




```
df["Order Date"] = pd.to_datetime(df["Order Date"])
df["Year_Month"] = df["Order Date"].dt.to_period("M")

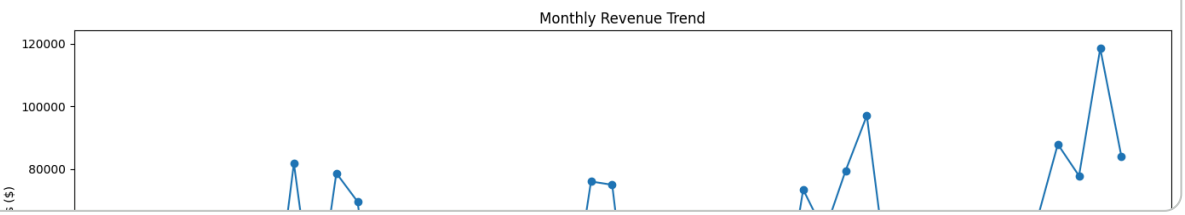
monthly_sales = (
    df.groupby("Year_Month")["Sales"]
    .sum()
)

plt.figure(figsize=(14, 6))

plt.plot(
    monthly_sales.index.astype(str),
    monthly_sales.values,
    marker="o"
)

plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales ($)")
plt.xticks(rotation=90)

plt.tight_layout()
plt.show()
```



```
category_profit = (
    df.groupby("Category")["Profit"]
    .sum()
    .sort_values(ascending=False)
)

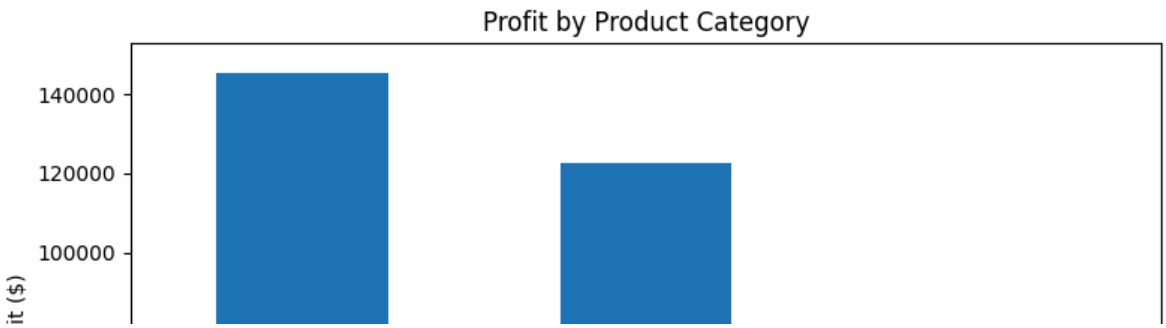
display(category_profit.reset_index())

plt.figure(figsize=(8, 5))
category_profit.plot(kind="bar")

plt.title("Profit by Product Category")
plt.xlabel("Category")
plt.ylabel("Total Profit ($)")
plt.axhline(y=0, linewidth=1)
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()
```

	Category	Profit
0	Technology	145454.9481
1	Office Supplies	122490.8008
2	Furniture	18451.2728




```
discount_analysis = (
    df.groupby("Discount")
    .agg(
        Total_Sales=("Sales", "sum"),
        Total_Profit=("Profit", "sum"),
        Average_Profit=("Profit", "mean")
    )
    .reset_index()
)

display(discount_analysis)

plt.figure(figsize=(9, 5))

sns.scatterplot(
    data=discount_analysis,
    x="Discount",
    y="Total_Profit"
)

plt.title("Discount vs Total Profit")
plt.xlabel("Discount")
plt.ylabel("Total Profit ($)")
```

```
plt.tight_layout()
plt.show()
```

	Discount	Total_Sales	Total_Profit	Average_Profit
0	0.00	1.087908e+06	320987.6032	66.900292
1	0.10	5.436935e+04	9029.1770	96.055074
2	0.15	2.755852e+04	1418.9915	27.288298
3	0.20	7.645944e+05	90337.3060	24.702572
4	0.30	1.032267e+05	-10369.2774	-45.679636
5	0.32	1.449346e+04	-2391.1377	-88.560656
6	0.40	1.164178e+05	-23057.0504	-111.927429
7	0.45	5.484974e+03	-2493.1111	-226.646464
8	0.50	5.891854e+04	-20506.4281	-310.703456
9	0.60	6.644700e+03	-5944.6552	-43.077212
10	0.70	4.062028e+04	-40075.3569	-95.874060
11	0.80	1.696376e+04	-30539.0392	-101.796797



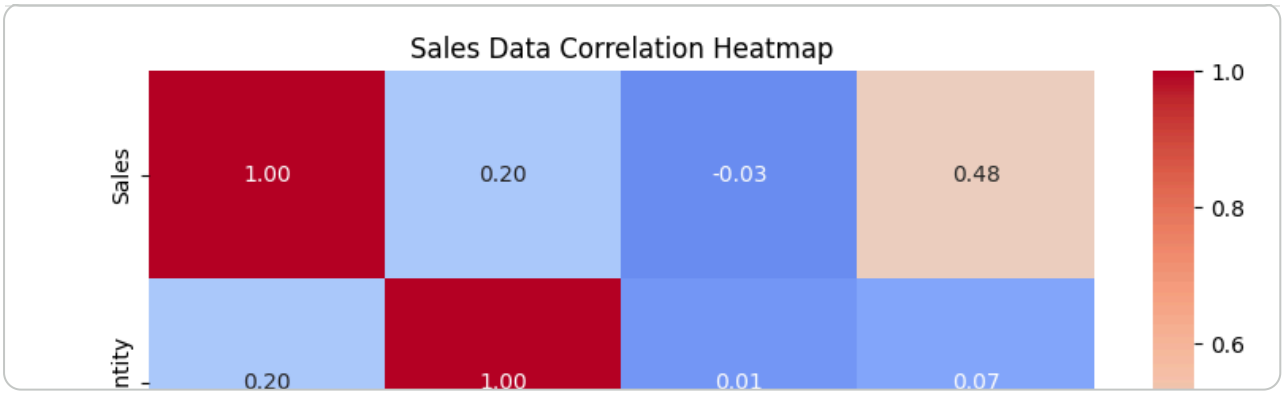

```
correlation_columns = [
    "Sales", "Quantity", "Discount", "Profit"
]

correlation = df[correlation_columns].corr()

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Sales Data Correlation Heatmap")
plt.tight_layout()
plt.show()
```




```
fig, axes = plt.subplots(
    2, 2, figsize=(16, 11)
)

# Monthly sales
axes[0, 0].plot(
    monthly_sales.index.astype(str),
    monthly_sales.values,
    marker="o"
)
axes[0, 0].set_title("Monthly Revenue Trend")
axes[0, 0].tick_params(
    axis="x", rotation=90
)

# Category sales
category_sales.sort_values().plot(
    kind="bar", ax=axes[0, 1]
)
axes[0, 1].set_title("Sales by Category")
axes[0, 1].tick_params(
    axis="x", rotation=0
)

# Regional sales
region_sales.sort_values().plot(
    kind="bar", ax=axes[1, 0]
)
axes[1, 0].set_title("Sales by Region")
axes[1, 0].tick_params(
    axis="x", rotation=0
)

# Top products
top_products.sort_values().plot(
    kind="barh", ax=axes[1, 1]
)
axes[1, 1].set_title("Top 10 Products by Revenue")

fig.suptitle(
    "Business Sales Performance Analytics Dashboard",
    fontsize=16
)

plt.tight_layout()

plt.savefig(
    "business_sales_dashboard.png",
    dpi=300,
    bbox_inches="tight"
)
```

```
plt.show()
```




```
print("========== BUSINESS INSIGHTS ==========")

print("Highest Revenue Category:",
      category_sales.idxmax())

print("Highest Revenue Region:",
      region_sales.idxmax())

print("Top Revenue Generating Product:"
```